

SUMIT DIXIT

C++ Developer & Tech Lead (Embedded)

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PROFILE

Seasoned software and embedded systems developer with over ten years of experience in C/C++ (11/14/17), Qt6/QML(5/6), Embedded C, and C#. Demonstrated proficiency in developing high-performance, secure, and reliable software and firmware solutions for telecom, networking, and defense sectors. Expertise includes real-time system optimization, microcontroller programming, signal processing, and cross-platform integration on Windows and Linux environments. Seeking a senior technical position in software development or embedded systems engineering, where I can apply my extensive knowledge in embedded systems, and protocol implementation. Aiming to contribute to advanced technological solutions in the technology.

WORK EXPERIENCE

Nov 2024 – Present
Tech Lead

Sopra Steria (AeroLine Project)

Project: NavBlue Pilot Application

Domain: Aerospace

Technologies: C/C++11/14/17, Qt 5/6, QML, CMake

Responsibilities

- Software Development: Designed and implemented new features based on customer requirements, ensuring their functionality through rigorous validation.
- Delivered production-ready builds for customer verification and deployment.
- Process Management: Applied SAFe (Scaled Agile Framework) methodology to streamline daily work processes. Utilized the Agility tool for project tracking and task management.
- Quality Assurance: Established and executed robust testing and validation procedures to maintain software quality and ensure compliance with aerospace standards.
- User Interface (UI) Development: Developed intuitive and user-friendly interfaces using QML to enhance functionality and usability.
- Technical Documentation: Authored detailed technical documentation and reports to support the software development lifecycle and regulatory compliance.

Jan 2024 – May-2024
C++ Senior Consultant

BRROYOFT (Client: Infosys – Boston Scientific)

Projects: Rezum/Calibrate/Diagnose Platform (Project: BSCFWSEZ [External Firmware])

Domain: Medical Device

Technologies: C/C++11/14/17, Qt 5/6, QML, CMake

Microcontroller : MUC/2

Responsibilities

- Led Development: Spearheaded the development of the Rezum generator software, utilizing C/C++11/14/17 and Qt 5/6 for creating advanced medical device solutions.
- Firmware Engineering: Designed and implemented firmware for the Rezum device, which uses RF energy to heat water into vapor for medical applications.
- Multithreading Applications: Developed robust multithreaded applications to manage high-speed data processing and control functions within the device.
- System Integration: Integrated software components to ensure seamless operation of the device, focusing on high reliability and efficiency.
- Requirements Gathering: Collaborated with stakeholders to gather and analyze requirements, ensuring the software meets clinical and technical needs.
- Quality Assurance: Implemented rigorous testing and validation processes to ensure software quality and compliance with medical device standards.
- User Interface Development: Developed user interfaces using QML to enhance the usability and functionality of the medical device.
- Technical Documentation: Created comprehensive technical documentation and reports to support the development lifecycle and regulatory submissions.

Apr 2022 – Oct 2023
Tech Lead

Fujitsu India Private Limited

Projects: FUJITSU Software PRIMECLUSTER

Domain: Storage Domain & Solutions

Customer: Nissan

Technologies: C/C++11/14/17, Python, Shell Script, Linux

Microcontroller: MUC/2

Responsibilities

- High-Availability Architecture: Designed and implemented a robust architecture for PRIMECLUSTER, ensuring maximum uptime for mission-critical systems.
- Optimization Strategies: Developed algorithms to optimize server utilization, improving overall system efficiency and performance.
- Targeted Failover Solutions: Created specialized failover mechanisms to quickly recover essential services, reducing the impact of system failures.
- Multi-Platform Integration: Integrated PRIMECLUSTER with various storage solutions, ensuring compatibility and seamless operation across different platforms.
- Advanced Scripting: Utilized Python and Shell Script for automation tasks, enhancing system monitoring and management capabilities.
- Security Enhancements: Implemented security protocols to protect sensitive data and maintain system integrity within the high-availability setup.
- Innovation in Resource Management: Introduced new methods for dynamic resource allocation, reducing operational costs and improving system responsiveness.
- Client-Specific Customization: Delivered tailored solutions to meet Nissan's specific storage and high-availability requirements, ensuring customer satisfaction and system reliability.

Nov 2020 – Apr 2022
Tech Lead

Larsen & Toubro Limited

Projects: Development of critical systems and equipment for oil & gas projects

Domain: Gas & Petrol

Customer: Halliburton

Technologies: MFC, VC++, C++11/14/17

Responsibilities

- Advanced System Design: Led the design and development of software for large tubular reactors and gas-to-liquid reactors, leveraging advanced C++11/14/17 technologies.
- Custom Equipment Integration: Integrated software solutions with L&T's proprietary equipment, ensuring seamless operation of test and production separators.
- Safety Compliance: Implemented rigorous safety protocols in software systems to comply with industry standards and ensure safe operation in hazardous environments.
- High-Performance Computing: Optimized software for high-performance computing, enabling efficient processing of complex simulations and calculations.
- Cross-Functional Collaboration: Collaborated with mechanical and chemical engineering teams to align software functionality with hardware specifications.
- Data Analytics Integration: Developed modules for real-time data analytics and monitoring, enhancing operational efficiency and predictive maintenance.
- Client-Focused Solutions: Delivered tailored software solutions for Halliburton, addressing specific needs in drilling and production processes.
- Innovation in Process Control: Pioneered new methods for process control automation, improving system accuracy and reducing manual intervention.

Feb 2020 – Jun 2020
Senior Consultant

Capgemini Technologies

Projects: Digital Transformation for McDonald's

Domain: Digital Retail Innovation

Customer: McDonald's

Technologies: C/C++11/14

Responsibilities

- Digital Retail Innovation: Led the development of software solutions to enhance the digital customer experience at McDonald's, focusing on innovative retail technologies.
- User Interface Design: Designed and developed intuitive and responsive user interfaces using Qt5, improving customer interactions and operational efficiency.
- Integration of Emerging Technologies: Implemented cutting-edge digital technologies to streamline restaurant operations and improve service delivery.
- Real-Time Data Processing: Developed systems for real-time data analytics to support dynamic decision-making and personalized customer experiences.
- Agile Development: Applied agile methodologies to ensure rapid development and deployment of features, meeting tight project deadlines.
- Global Digital Retail Center: Contributed to the establishment of Capgemini's Global Digital Retail Center, focusing on showcasing business innovation and attracting talent.
- Cross-Platform Compatibility: Ensured software solutions were compatible across various platforms, enhancing flexibility and scalability.
- Stakeholder Engagement: Collaborated with McDonald's stakeholders to gather requirements and provide tailored digital solutions, aligning with their strategic goals.

Aug 2014 – Feb 2020
Senior Research Developer

KeySight Technologies International Private Limited

Domain: Telecom & Networking

Customer: Defense Organization

Technologies: C/C++11/14, Qt5/QML, Windows/Linux Platform

Responsibilities

- **SCPI Parser Development:** Led the design and implementation of an SCPI parser, enabling precise communication and control of test and measurement instruments in defense applications.
- **Cross-Platform Development:** Developed and maintained software solutions that operated seamlessly across Windows and Linux platforms, ensuring versatility and wide adoption.
- **Real-Time Data Acquisition:** Engineered systems for real-time data acquisition and processing, crucial for mission-critical defense communications.
- **Protocol Integration:** Integrated custom communication protocols to support specific requirements of defense networking systems.
- **High-Performance Optimization:** Optimized software for high-performance computing, enabling rapid data analysis and efficient resource utilization.
- **GUI Development:** Created user-friendly graphical user interfaces (GUIs) using Qt5/QML, enhancing usability and operator efficiency.
- **Security and Compliance:** Implemented security features and compliance protocols to protect sensitive data and ensure adherence to defense standards.
- **Custom Solution Design:** Collaborated with defense clients to design custom software solutions that met specific operational needs, improving functionality and reliability in complex environments.

Jun 2010 – Jan 2014
Research & Developer

Agilent Technologies Private Limited

Domain: Telecom & Networking

Customer: Defense Organization

Technologies: Embedded C, C# (Framework 4.5)

Microcontroller: MUC/2

Responsibilities

- **Firmware Development:** Created and maintained embedded firmware using Embedded C for MUC/2 microcontrollers, tailored to telecom and networking systems.
- **Signal Processing Algorithms:** Developed advanced signal processing algorithms to enhance the functionality and efficiency of telecom equipment in defense applications.
- **Embedded C Optimization:** Optimized embedded C code for performance, ensuring efficient use of limited hardware resources in microcontroller environments.
- **C# Application Interfaces:** Designed and developed C# applications to provide user interfaces for configuration and monitoring of embedded systems.
- **Microcontroller Programming:** Programmed and debugged microcontrollers to ensure accurate and reliable operation in complex telecom systems.
- **Defense-Specific Features:** Implemented features specific to defense requirements, such as secure communication protocols and data encryption.
- **Power Management Solutions:** Developed power management techniques to optimize energy consumption in battery-powered defense devices.
- **Environmental Testing:** Conducted environmental testing to validate system performance under various conditions, including temperature extremes and electromagnetic interference.

EDUCATION

Bachelor of Engineering in UPTECH University, 2008

Graduated with 68.9%

CERTIFICATIONS

- IITK AIML Programming Python • IITK AIML: Data Science with Python
- IITK AIML: Machine Learning • Qt6 GUI Development

SOFTWARE PROFICIENCY

- **Programming Languages:** C/C++11/14/17, VC++ (MFC), Python, Perl/Shell, SQL
- **Frameworks & Libraries:** Qt6/QML5, MFC, Win32, .NET
- **Technologies & Concepts:** Object-Oriented Programming (OOPS), Object-Oriented Design (OOD), Multithreading, Data Structures, UML, SOLID Principles, Design Patterns
- **Operating Systems:** Windows 10/11, Linux (RedHat 8.1, 7.4)
- **Build & Configuration Tools:** Bitbucket, GIT, Jira
- **Cloud Computing:** AWS

SKILLS

- Project Management
 - Team Leadership
 - Requirements Analysis
 - System Design
 - Solution Architecture
 - Cross-Functional Collaboration
 - Technical Documentation
 - Quality Assurance
- Customer Engagement
 - Agile Methodologies
 - Unit Testing
 - Performance Optimization
 - Risk Assessment
 - Vendor Coordination
 - Process Improvement
 - Technical Training and Mentoring

LANGUAGE PROFICIENCY

Language: English	Language: Japanese	Language: German	Language: German
Level: Advanced	Level: Basic	Level: A2	Level: B1
Institute:	Institute: JLPT	Institute: Goethe	Institute: Goethe (Running)